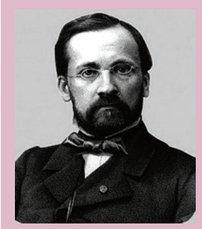
DO YOU KNOW?

The process of vaccination was introduced by Edward Jenner. According to the World Health Organisation (WHO), Jennerian vaccination has eliminated small pox totally from the human population.

Killed Vaccines: Micro organisms (bacteria or virus) killed by heat or chemicals are called killed or inactivated vaccines. They require a primary dose followed by a subsequent booster dose. e.g. Typhoid vaccine, cholera vaccine, pertussis vaccine.

Know your Scientist



Louis Pasteur is an 18th century French chemist and microbiologist. He coined the term vaccine. Pasteur developed vaccine against chicken pox, cholera, anthrax, etc.

22.9.2 Immunization Schedule

The World Health Organization in the year 1970 has given a schedule of immunization for children. This schedule is carried out in almost all countries. Table 22.8 gives the schedule of vaccination procedures followed in India.

BCG (Bacillus Calmette Guerin): This was prepared by two French workers Calmette and Guerin (1908-1921). The bacilli are weakened and used for immunization against tuberculosis.

DPT (Triple Vaccine): It is a combined vaccine for protection against Dipetheria, Pertussis (whooping cough) and Tetanus.

MMR: Mumps, Measles, Rubella vaccine gives protection against viral infections.

DT: It is a dual antigen or combined antigen. It gives protection from Diphtheria and Tetanus.

TT (Tetanus Toxoid): Toxin of Tetanus bacteria.

TAB: Combined vaccine for typhoid, paratyphi A and paratyphi B.

Table 22.8 Immunization Schedule

Age	Vaccine	Dosage
New born	BCG	1st dose
15 days	Oral Polio	1st dose
6th week	DPT and Polio	1st dose
10th week	DPT and Polio	1st dose
14th week	DPT and Polio	1st dose
9 – 12 months	Measles	1st dose
18 – 24 months	DPT and Polio	1st dose
15 months – 2 years	MMR	1st dose
2 – 3 years	TAB	2 doses at 1 month gap
4 – 6 years	DT and Polio	2nd booster
10th year	TT and TAB	1st dose
16th year	TT and TAB	2nd booster

Activity 3

Recently in 2018, Nipah virus was in the headlines of the daily newspaper. Collect the following information. What is Nipah virus? How it gets transmitted? Mention the preventive measures taken by the Government to check the disease.

Points to Remember

- Bacteria are single celled prokaryotic organisms, without a well defined nucleus (nucleoid) and other cell organelles. The genetic material is DNA.
- Viruses are small microscopic infectious agents that can multiply only inside the living cells.
- Fungi are group of eukaryotic heterotrophs which are either single celled (Yeast) or multicellular (*Penicillium*, *Agaricus*).
- Microorganisms which enrich the soil with nutrients are called as biofertilizers.

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